Light reflection rheography: A simple noninvasive screening test for deep vein thrombosis

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Purpose: **The clinical diagnosis of deep vein thrombosis (DVT) is unreliable. Contrast phlebography (CP) continues to be the gold standard, but it is invasive. Although duplex ultrasonography is an accurate, noninvasive alternative, it is expensive, technically demanding, and time-consuming. We postulated that light reflection rheography (LRR), a noninvasive method of assessing the quantity and rate of venous emptying, might be a reliable and inexpensive bedside approach to screening patients with clinically suspected DVT.**

Methods: **With LRR, infrared light is beamed onto the skin, and the amount of backscattered rays are detected, which indirectly measures the amount of blood present in a volume of the epidermis beneath the LRR probe. Applied to the calf muscle pump, LRR can provide a noninvasive method of assessing blood volume changes in the sample area of skin, in response to venous hemodynamic changes in the lower limb.**

Results: **Sixty-nine limbs in 61 patients undergoing CP for clinically suspected DVT over a period of 12 months also underwent LRR, either just before or within 24 hours of undergoing phlebography. The criteria for diagnosing DVT on CP were presence of filling defect or nonfilling of a venous segment. The result of LRR was considered positive for DVT if the rate of venous emptying was 0.35 or less. With these criteria a sensitivity of 96.4% and specificity of 82.9 were obtained. This resulted in a positive predictive value of 79% and a negative predictive value of 97.1%.**

Conclusions: **LRR is a simple, inexpensive, and noninvasive bedside test that takes 10 minutes to perform. It is highly sensitive with a high negative predictive value, detecting most cases of DVT, reliably ruling out DVT, and eliminating the need for more time-consuming and costly studies. Therefore it seems to be an appropriate screening test in patients with clinically suspected DVT. (J VASC SURG 1993;18:767-72.)**

Clinical diagnosis of deep vein thrombosis (DVT) in the lower limb remains inaccurate, being correct on average about 50% of the time.[1] Most patients with DVT remain symptom free, and physical signs, when present, are unreliable.[2] Because DVT can lead to potentially fatal complications, it is important that clinical suspicion be confirmed by an objective, preferably noninvasive test. Until recently, contrast venography was the gold standard for diagnosing DVT, despite its invasiveness and associated morbidity rate.[3] Impedance plethysmography (IPG) has been investigated widely and is sensitive in detecting occluding thrombi in the iliac, femoral, and popliteal veins, but it is insensitive in detecting calf vein thrombosis.[4,5]

Real-time B-mode ultrasonography (duplex) is now the accepted gold standard for the diagnosis of proximal DVT in many centers because its accuracy virtually matches that of venography.[6] These techniques, although safe, effective, and accurate, have not been widely used for screening purposes because they require considerable operator expertise, both in performing the test and in interpreting the results.

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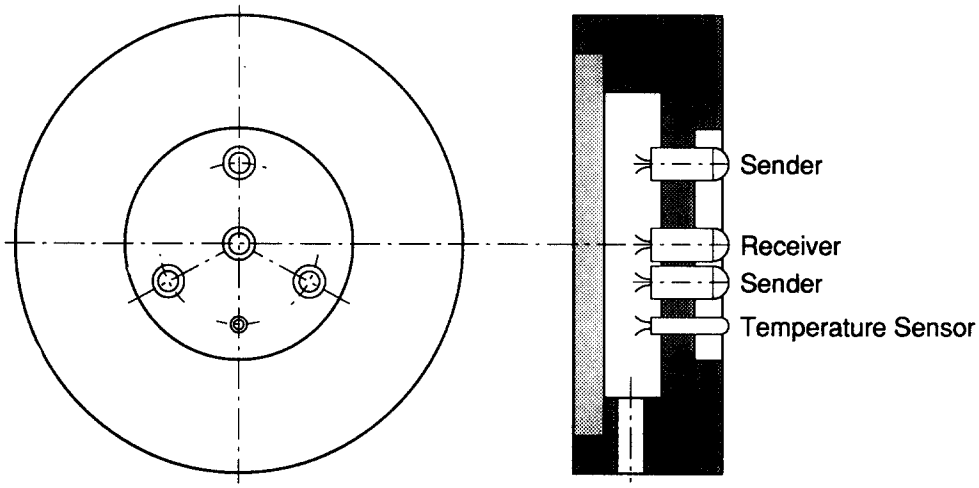


Fig. 1. LRR probe head.

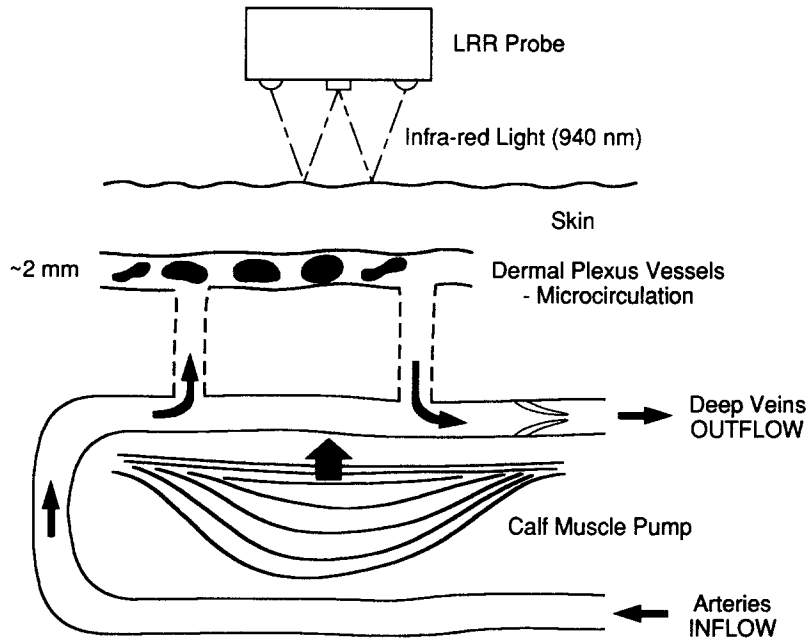


Fig. 2. Mechanism by which LRR trace is generated.

Duplex ultrasonography and IPG can be time-consuming, thus requiring considerable patient cooperation; they are expensive and insensitive in isolated calf DVT, and they generally require a hospital setting.

Light reflection rheography (LRR) is a refined form of photoplethysmography (PPG).[7] It uses infrared light of 940 nm wavelength, which is impervious to skin pigmentation, and a redesigned probe that incorporates three light-emitting diodes instead of an electrode. This provides homogeneous illumination of the skin dermal plexus as opposed to just reflection from the topmost capillary surfaces. LRR does not require operator calibration and correlates with the venous pressure changes associated with exercise measured by invasive means, by measuring infrared light reflected from the dermal microcirculation.[8] The use of LRR has been investigated in chronic venous insufficiency, but there have been only sporadic reports claiming the usefulness of LRR in the investigation of patients with DVT.[9-13]

The purpose of this study was to determine the

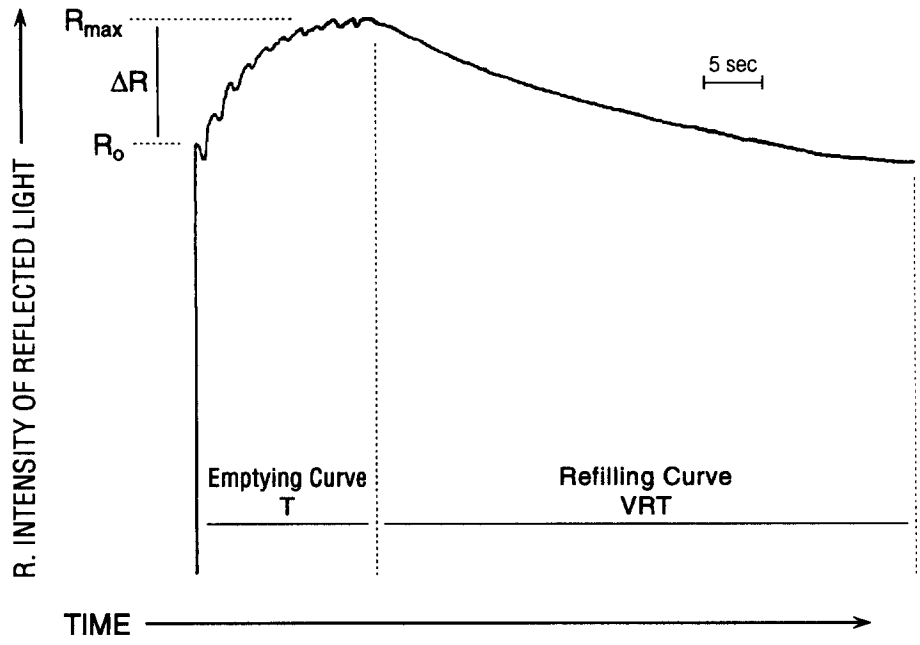


Fig. 3. LRR trace.

accuracy of LRR in the diagnosis of DVT compared with contrast venography as the gold standard.

PATIENTS AND METHODS

All patients suspected of having DVT on clinical examination and referred for contrast venography were included in the study. Patients referred for duplex scanning were not included in the study because venography was used as the gold standard. There were no standard criteria for clinical diagnosis because the patients came from different departments in the hospital, including the emergency department.

LRR was performed either before or after venography, but always within 24 hours, by one of the authors, either in the radiology department, the emergency department, or on the ward as a bedside procedure. Both procedures were performed during the same visit. If LRR was performed after venography, results of venography were not known to the operator. Results from both LRR and venography were reported independently, the venograms by a consultant radiologist and the LRR result by the author performing the test. The results of LRR and venography were then compared.

LRR was performed with a portable device (A-V1000; Hemodynamics, Inc., Boca Raton, Fla.). This consists of a probe and two combined elements: an electronic measuring and evaluation unit and a recorder. The current is supplied by built-in rechargeable batteries so that the apparatus can be easily moved. The LRR probe consists of three infrared light-emitting diodes arranged circumferentially around a single photodetector. It is also equipped with a small thermocouple for the measurement of skin temperature (Fig. 1). Light is beamed vertically onto the dermal vessel plexus. Having penetrated 1 to 2 mm into the skin (Fig. 2), some of it is absorbed by the erythrocytes, and the remainder is backscattered and picked up by the photodetector, which is converted to an electrical signal and recorded.

For the actual test, the patient was seated in a chair with the hip extended approximately 110 degrees and the feet flat on the floor, with the knee also extended to approximately 110 degrees which provided a maximum range of dorsiflexion at the ankle. The skin was cleaned, and the probe was attached to the skin about 10 cm above the medial malleolus with a double-sided adhesive ring. The machine was allowed to calibrate and the reading on it was allowed to stabilize. When a steady signal Ro was registered, (Fig. 3) the patient was required to perform 10 dorsiflexions over a 15-second period, in time to a built-in metronome, with both an auditory and a light signal. The test was repeated after a 2- to 3-minute rest period. The curve with a better venous emptying was used for analysis. After this the patient relaxed and remained steady until the signal trace returned to normal in 40 seconds. The shape of the LRR curve is a function of the calf muscle pump. During dorsiflexion, the muscle pump is activated

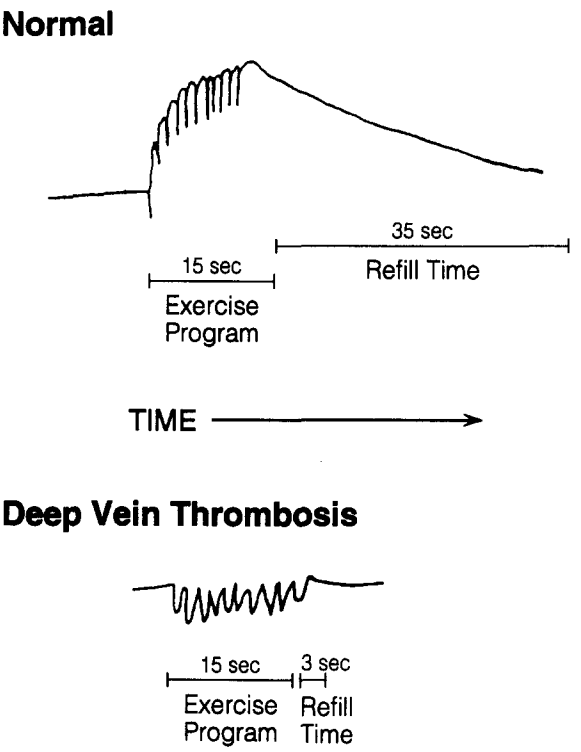


Fig. 4. Source of LRR traces that can be generated.

and causes a drop in the venous pressure because of blood being ejected into the proximal deep veins of the leg. Because the venous blood is drained from the surface into the deep vessels, the dermal vessel plexus is emptied. The LRR trace shows this as an increase in the amount of reflected light (R) because there is less blood to absorb the incoming infrared light. At the end of the exercise, venous emptying and light reflectivity is maximal (Rmax). After the exercise the trace returns to a baseline value as the veins refill from arterial in-flow, yielding a resting value in the vessel plexus. This time period is the venous refill time, and the time required to reach Rmax is the emptying time (T seconds). The change in light reflection caused by the exercise is called Delta R (ΔR) and the rate of change of light reflection (Δ R/T) was calculated in each case (Fig. 3). In the presence of occluding DVT a flat LRR curve is registered (Fig. 4).

Standard ascending venography was performed in all cases with the Lee Thomas technique.[14] The venograms were read by a consultant radiologist. The criteria for DVT was either an intraluminal filling defect or nonfilling of a venous segment. The level of the thrombosis was interpreted as either limited to the calf only, extending proximally into the popliteal, femoral, and iliac vein; or limited to the proximal vessels only.

Table I. Comparison of LRR with contrast venography

	Venography	
LRR	*Positive*	*Negative*
Positive	27	7
Negative	1	34

LRR sensitivity, 96%; specificity, 83%; PPV, 79%; NPV, 97%.

RESULTS

Sixty-nine limbs in 61 patients undergoing contrast venography for clinically suspected DVT were included in the study. The ages of the patients ranged from 22 to 82 years (mean 62). The study population consisted of 39 women and 22 men entered over a period of 12 months.(January 1991 to December 1991)

Venography detected DVT in 28 (40%) limbs, of which 16 DVTs occurred in patients on geriatric wards, six occurred in postoperative patients, and six occurred in patients from the emergency department. Five (18%) cases of DVT were limited to the calf only. In 14 (50%) limbs the thrombosis extended into the proximal veins, and in nine (32%) limbs the thrombosis was limited to the proximal veins. There was no evidence of DVT in 41 (60%) limbs.

The LRR curve in DVT is a flat curve with minimal or no venous emptying and a minimal refill time (Fig. 4). The rate of venous emptying on the LRR curve, Δ R/T was the parameter used to diagnose DVT as recommended by Mitrani et al.[13] A value of Δ R/T ≤ 0.35mm/sec was considered positive for DVT (Fig. 3). With the above criteria LRR correctly diagnosed DVT in 27 of 28 venographically diagnosed cases (96% sensitivity) (Table I). LRR also identified 34 of 41 limbs that were negative for DVT on venography (83% specificity). There were seven false-positive results (positive predictive value [PPV] 79%) and one false-negative result (negative predictive value [NPP] 97%). LRR detected all cases of thrombosis limited to the calf and missed one case of thrombosis in the proximal veins. The venograms of the five cases with thrombosis limited to calf veins showed extensive multisegmental calf vein DVT extending to the popliteal vein in four cases and limited to calf veins only in one. The hemodynamic measurements in these five cases were not any different from the measurements for more proximal DVT.

DISCUSSION

DVT is a fairly common condition, especially among hospital patients and is associated with considerable morbidity and potentially lethal complications. A high index of suspicion is therefore necessary for the clinical diagnosis of DVT, which unfortunately is unreliable because its accuracy is in the range of 30% to 50%.[1] This results in the performance of a large number of unnecessary diagnostic tests such as venography. Venography is invasive, is associated with contrast medium–related complications, is expensive, and sometimes is extremely difficult to perform or interpret.[14] In our study 60% of the venography results were negative and could have theoretically been avoided if a simple noninvasive, objective screening test would have been used to rule out DVT. Duplex ultrasonography, although noninvasive and accurate for proximal DVT, requires considerable operator expertise, can be time-consuming, and is expensive. It is usually not available in the emergency department setting, and the sensitivity of duplex ultrasonography drops to 50% to 70% below the level of the knee.[15] Although the management of DVT localized to the calf is controversial, approximately 20% of these thrombi propagate proximally and then have the potential risk of embolization.[16]

IPG is again a sensitive noninvasive tool for diagnosing proximal DVT, but is not sensitive when it comes to thrombosis limited to the calf veins.[4,5] IPG also requires operator expertise, can be time-consuming, and requires considerable patient compliance, especially when serial IPGs are to be performed. It has been largely replaced by duplex ultrasonography. PPG provides an accurate measure of venous refill time associated with calf exercise, but it is unable to quantify venous emptying without cumbersome operator calibration. LRR is basically quantitative PPG and can quantify both parameters of venous function, namely, venous emptying and venous refill time and in our study identified all five cases of calf DVT. Its sensitivity (96%) was about the same regardless of the location of the thrombus.

One limb with thrombus in the popliteal and superficial femoral vein was not detected by LRR. We believe this was due to the fact that the thrombus was about 4 to 7 days old and that several collateral vessels were seen on the venogram, which probably provided alternative outflow channels. There were seven false-positive results on LRR. Three of these patients had severe edema of the ankle and foot, which made a full range of dorsiflexion at the ankle difficult. In two of the patients pain caused by cellulitis prevented good dorsiflexions at the ankle. In two patients we could not find any explanation. Therefore patients with severe pedal and ankle edema and painful conditions of the foot and ankle are not suitable candidates for LRR. Patients with casts that prevent the placement of the LRR probe and immobile, bedridden patients are not suitable candidates for LRR. Patients with chronic venous insufficiency would theoretically have grossly abnormal (short) venous refilling time.

The PPV of LRR was 79%, but it is better to err on the positive side because a more definitive test can then confirm or refute the diagnosis. LRR achieved a high NPV of 97%, which makes it very suitable for a first line screening test to "rule out DVT" and thereby bring about a reduction in a great number of unnecessary venograms or duplex scans. When DVT is suspected strongly on clinical grounds and the results of duplex scanning or IPG are negative, one would normally repeat the test in 24 to 48 hours and maybe even later. Serial IPGs and duplex scanning require considerable patient compliance and can be expensive. A duplex ultrasound study, which is most commonly performed, costs approximately $350.00 at our institution and the cost of contrast venography is approximately $800.00 to $900.00. An LRR study on the other hand costs $70.00 to $90.00. LRR as a first line screening test therefore would save a tremendous amount of money considering that nearly 50% of the results of venography and duplex scanning for DVT are negative. LRR is extremely useful in the emergency department and "out of hours" setting because it is easy and quick to perform. It is not operator dependent and can be performed by a nurse or an intern as a bedside procedure with minimal training.

In conclusion, LRR seems to be an appropriate screening test for DVT. It is highly sensitive with a high NPV, noninvasive, inexpensive (cost of equipment $3,900), easy to perform, and interpret, portable and can be performed as a bedside procedure and provides one with a hard copy of the result. A negative LRR result safely "rules out" DVT but a positive result mandates a definitive test, either a duplex ultrasound scan or venography. LRR may emerge as one of the more sensitive noninvasive tests for diagnosing isolated calf DVT.

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